

Power and Bandwidth Allocation for Multi-Target Tracking in Collocated MIMO Radar

Haowei Zhang , Binfeng Zong, and Junwei Xie

Abstract—A collocated multiple-input and multiple-output (MIMO) radar has the ability to track multiple targets under the simultaneous multi-beam mode, where each transmit beam is defocused to search the entire surveillance space, and then; a focused beam is formed at the receiver to achieve a higher resolution. A joint power and bandwidth allocation strategy is developed in this mode. The optimization model is established with the aim of improving the sum of weighted posterior Cramer-Rao lower bounds (PCRLBs) of multiple targets, which is predicted based on the feedback information in the tracking recursive cycle, subject to the limited power and bandwidth budget. By exploiting the convex relaxation technique and the cycle minimizer, this problem is transformed into a two-stage convex problem, and an efficient solution is provided using the cyclical optimization framework. The effectiveness of the proposed strategy is demonstrated by comparison with three uniform resource allocation methods in many cases.

Index Terms—Collocated MIMO radar, resource allocation, multi-target tracking, PCRLB, nonconvex optimization.

I. INTRODUCTION

THE MIMO radar possesses a large number of benefits over traditional phased array radars, such as path diversity [1], simultaneous multi-function and flexible beam pattern [2]. As such, it has been receiving much attention in recent years.

Generally, the MIMO radar system can be divided into two types according to the relative distance between arrays: the collocated MIMO radar [3] and the distributed MIMO radar [4]. The former one is able to synthesize various beam patterns by taking advantage of waveform diversity, i.e., the multiple uncorrelated/orthogonal waveforms, while the later type enjoys the spatial diversity owing to viewing targets from different aspects. In order to make full use of these advantages, the finite resources in radar systems need to be efficiently managed.

The resource management in multi-user MIMO (MU-MIMO) communication and MIMO radar systems has been investigated in many works. The energy efficiency is a classical issue among them. In the MU-MIMO communications, [5] tries to optimize the transmit covariance through the information theory, and the subgradient method and Dinkelbach method with the bisection search method are used to construct the iterative algorithms.

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Yu *et al.* [6] proposes three computationally efficient power allocation schemes for the downlink of a MU-MIMO system with beamforming over composite Rayleigh channels. The gradient projection method and the Karush–Kuhn–Tucker (KKT) conditions are exploited to achieve the optimality. Wu *et al.* [7] puts forward a two-stage power allocation strategy incorporating with the spectrum efficiency, where the user priorities and user grouping method are used to satisfy the user equipment requirements. In [8], an energy and spectrum efficiency tradeoff-based objective is considered for the primary users and the secondary users in radio networks. A damped three dimensional (D3D) message-passing algorithm (MPA) based on deep learning is proposed to solve such an NP-hard. The joint power, original bandwidth, and detected hole bandwidth allocation for multi-homing heterogeneous networks is investigated in [9], and the continuity relaxation is adopted to transform the mixed integer nonlinear optimization problem into a convex one.

In MIMO radars, the targets to be detected, localized, or tracked, can be regarded as users in MU-MIMO communications. However, unlike that the MU-MIMO communications just convey the information from one user to others, most of radars actively transmit electromagnetic waves and receive the backscatters from the targets, to acquire their information. Therefore, the optimization criterions of resource allocation between MU-MIMO communications and MIMO radars are different. In the former systems, the energy efficiency, satisfaction to user equipment, Quality of Service (QoS), and fairness are usually adopted to perform this task, while the probability of detection [10], [11], the mutual information [12], [13], the Cramer-Rao lower bound (CRLB) [14]–[20], and the posterior CRLB (PCRLB) [21]–[25], which quantifies the performance of target detection [10], [11] or state estimate [12]–[25], are always used in radars. In [10], [11], the game theory is adopted to address the power allocation in a multistatic MIMO radar network, and there is no communication between each cluster. Based on the information theory, [12], [13] use the KKT-based, the sequential optimization-based and three-step-based algorithms to study the power allocation issues in MIMO radar and communication coexistence systems. In the distributed MIMO radars, [14], [15] propose the multi-start local search algorithm to activate a subset of arrays and to minimize the power consumption. Alternatively, [16], [17] establish the unified power optimization models, and solves them through the relaxation with the domain decomposition method and the second-order cone program-based algorithms, respectively. To improve the solution efficiency in [16], two iterative decomposition algorithms and the

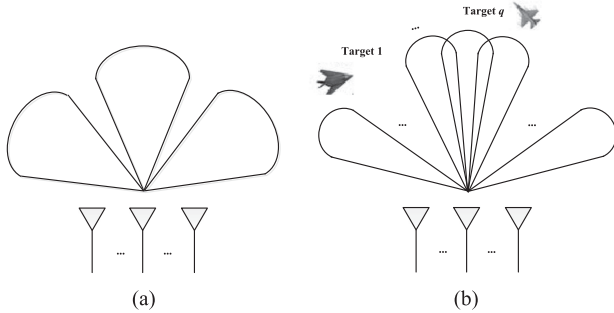


Fig. 1. Illustration of multi-target tracking by simultaneous multi-beam mode. (a) Transmit beam pattern. (b) Receive beam pattern.

alternating global search algorithm are respectively proposed in [18] and [19]. Considering the probabilistic uncertainty of target reflectivity, [20] formulates the power allocation as the chance constrained programming model, and transforms it into the deterministic convex one. *However, these works [10]–[20] are aiming at static targets. In reality, the dynamic property of targets should be highlighted.* Thus, the study of resource allocation in target tracking is more practical.

The collocated MIMO radar is capable of launching multiple, simultaneous and orthogonal beams, and each beam can execute tasks independently. This characteristic is very effective in the multi-target tracking, since the peak power is reduced to meet the requirement of low probability of interception; meanwhile, the dwell time on each target is prolonged so that a higher Doppler resolution is attained [21]. When transmit beams are widely distributed in the entire surveillance area, the focused beams are formed at the receiver to cover the corresponding region. This mode enables the spatial diversity information of targets can be exploited to distinguish the receive signals. As such, the multi-target tracking problem can be divided into several single target tracking problems. The illustration of simultaneous multi-beam mode is shown in Fig. 1. (Note that there are no real beams in the space due to the orthogonality, and the equivalent transmit beams and receive beams are all synthesized at the receiver.) Refs. [21]–[25] study the power allocation problem in this mode, and the convex/modified convex optimization methods are applied for the solution. One choice to achieve the orthogonality is to assign the discrete bandwidth to each transmit beam. Such a scheme shows more advantages over other orthogonal beam-forming techniques [26], [27], since the waveform diversity can be easily achieved by simple filtering and the bandwidth of each transmitter can be flexibly changed without breaking the orthogonality. *However, the works above [10]–[25] all ignore the bandwidth influence.* Similar with MU-MIMO systems, the bandwidth is also a crucial resource in practical applications. In the modern air defense wars, a collocated MIMO radar usually needs to perform multiple tasks/functions simultaneously, e.g., surveillance, multi-target tracking, and communications, and the execution of multiple tasks easily results in the limitation of bandwidth. However, to our best knowledge, the joint power and bandwidth allocation in collocated MIMO radars is not yet available so far. Since [21]–[25] have proposed the power

allocation scheme for multi-target tracking; we are going to ask 1) *which is the more critical resource between power and bandwidth, and 2) whether the tracking performance can be further improved by the joint power and bandwidth allocation.*

It has been shown that the optimization criterion in target tracking, i.e., the PCRLB, is a convex function in terms of the power variable [23]. However, the problem becomes dubious after the bandwidth being incorporated. In addition, unlike the MU-MIMO communication systems, where the power and spectrum can be separately considered and optimized [9], the power and the bandwidth are coupled in the PCRLB. The above two factors lead to an intractable problem that can't be solved via a single algorithm. Aiming at the real-time demand in radar systems, an efficient solver is proposed.

The cognition idea [28] is inspired by the predation procedure of the bat, when the bat first perceives the environment and then acts based on the information it has perceived. With this motivation, the author in [28] proposes the conception of cognitive radar/radio system, which can perceive the environment change and adapt its parameters to the change. For the cognition radar, three salient features are identified: Bayesian inference, which ensures the environment perception; feedback from the receiver to the transmitter, which enables the intelligent adjustment of parameters; and the adaptive processing at the transmitter, which facilitates the radar learning from the environment. In the target tracking cycle, the target posterior state estimate can be used as the prior information at next time. Thus, we use the cognition idea to form a closed loop tracking system, where the power and the bandwidth are adaptively allocated according to the target state change.

The main contributions of this paper are summarized as follows.

- 1) The joint power and bandwidth allocation problem is established as an optimization model subject to the given resource budget. Since the PCRLB [29] bounds the target state estimate and is predictive, it is derived and adopted as the optimization criterion. Previous works adopt the sum or the maximum (worst case) of PCRLB [21]–[24] as the objective function, which may be oversimplified. To describe the specific requirement of tracking accuracy of each target, a weight factor is introduced and the sum of weighted PCRLBs is used to perform this task.
- 2) An efficient algorithm which combines the convex relaxation and the cyclical minimizer [30] is proposed for solving the optimization model. This algorithm relaxes the constraint of one variable on the condition of another variable being fixed, and optimizes the two variables in a cyclical manner. Simulation results show the effectiveness and efficiency of proposed solver compared with the genetic algorithm [31].
- 3) A closed loop feedback scheme for the power and bandwidth allocation is developed. We use the square-root cubature Kalman filter (SCKF) [32], [33], which has a simple structure but high accuracy, to obtain accurate target state estimates under the background of the nonlinear measurement. The filtering results at the current time are used for the PCRLB prediction and guiding the resource

TABLE I
 NOTATIONS AND EXPLANATIONS

superscript q	q -th target, $q \in \{1, \dots, Q\}$
subscript k	k -th sample/tracking interval
$\mathbf{x}_k^q$	Target state vector, which describes the target motion
P_k^q	Allocated power, which is contained in the transmit signal
β_k^q	Allocated bandwidth, which is contained in the transmit signal
τ_k^q	Time delay, which indicates the target range
ν_k^q	Doppler shift, which indicates the target radial velocity
$\mathbf{z}_k^q$	Measurement vector, which contains the target information extracted from the receive signal

allocation at the next time. Then, the optimal resource allocation achieved by the proposed solver renders more accurate target tracking, and a virtuous circle loop is realized.

- 4) Through the simulations in many cases, the advantages of our strategy are demonstrated in comparison with three uniform resource allocation methods.

The rest of the paper is organized as follows. The system model is established in Section II. Section III derives the PCRLB in terms of the power and bandwidth. In Section IV, the optimization model is formulated and an efficient solver is proposed, based on which the closed loop feedback scheme for resource allocation is built. In Section V, numerical results are presented to verify the superiority of the proposed method. Section VI concludes this article.

II. SYSTEM MODEL

For clarity, some notations used in this section are listed in Table I, and others will be explained at which they first appear.

A. Signal Model

Assume that a 2-D monostatic collocated MIMO radar tracks multiple targets with the simultaneous multi-beam mode. Due to the targets being widely separated in the surveillance region, each beam is only responsible for one target (see Fig. 1). Thus, in the narrowband situation, the transmit waveform for the q -th target at the k -th sample interval is [21]

$$s_{t,k}^q(t) = \sqrt{P_k^q} \tilde{s}_k^q(t) \exp(-j2\pi f_c t), \quad (1)$$

where $q = 1, 2, \dots, Q$. $\tilde{s}_k^q(t)$ is the complex envelope of the transmit signal, which implies two important factors: the effective bandwidth [16]

$$|\beta_k^q|^2 = \frac{\int f^2 |\tilde{s}_k^q(f)|^2 df}{\int |\tilde{s}_k^q(f)|^2 df}, \quad (2)$$

and the effective time duration

$$|T_k^q|^2 = \frac{\int t^2 |\tilde{s}_k^q(t)|^2 dt}{\int |\tilde{s}_k^q(t)|^2 dt}. \quad (3)$$

where f and t in (2) and (3) represent the frequency and time domain, respectively. β_k^q and T_k^q both have effects on the tracking accuracy, which will be shown in the following parts. f_c is the carrier frequency. The receive waveform is given by

$$s_{r,k}^q(t) \approx h_k^q \sqrt{\alpha_k^q P_k^q} \tilde{s}_k^q(t - \tau_k^q) \exp(-j2\pi \nu_k^q t) + n_k^q(t), \quad (4)$$

where h_k^q is reflection coefficient, and α_k^q is the propagation attenuation caused by the path loss. τ_k^q and ν_k^q are time delay and Doppler shift, respectively. $n_k^q(t)$ is the zero mean, complex white Gaussian noise.

B. Target State Model and Measurement Model

For simplicity, we assume that the target motion follows the constant acceleration (CA) mode [34]

$$\mathbf{x}_k^q = \mathbf{F}_q \mathbf{x}_{k-1}^q + \mathbf{w}_{k-1}^q, \quad (5)$$

where $\mathbf{x}_k = [x_k, y_k, \dot{x}_k, \dot{y}_k, \ddot{x}_k, \ddot{y}_k]^T$ denotes the target state, and (x_k, y_k) , $(\dot{x}_k, \dot{y}_k)$, and $(\ddot{x}_k, \ddot{y}_k)$ denote the range, velocity and acceleration, respectively. $\mathbf{w}_{k-1}^q \sim \mathcal{N}(\mathbf{0}, \mathbf{Q}_q)$, where $\mathcal{N}(\mathbf{0}, \mathbf{Q}_q)$ denotes the Gaussian distribution with zero-mean and covariance matrix $\mathbf{Q}_q$. The transition matrix $\mathbf{F}_q$ and the covariance matrix of process noise $\mathbf{Q}_q$ are

$$\mathbf{F}_q = \begin{bmatrix} 1 & T_s & T_s^2/2 \\ 0 & 1 & T_s \\ 0 & 0 & 1 \end{bmatrix} \otimes \mathbf{I}_2, \quad (6)$$

and

$$\mathbf{Q}_q = \kappa_q \begin{bmatrix} T_s^5/20 & T_s^4/8 & T_s^3/6 \\ T_s^4/8 & T_s^3/3 & T_s^2/2 \\ T_s^3/6 & T_s^2/2 & T_s \end{bmatrix} \otimes \mathbf{I}_2, \quad (7)$$

respectively. $\mathbf{I}_2$ denotes the 2×2 identity matrix, $\otimes$ denotes the Kronecker operator, T_s is the sample interval, and κ_q is the process noise intensity.

The nonlinear measurement model of the q -th target $\mathbf{z}_k^q$ is described as

$$\mathbf{z}_k^q = h(\mathbf{x}_k^q) + \mathbf{v}_k^q, \quad (8)$$

where $h(\cdot)$ is the nonlinear transformation, with

$$h(\mathbf{x}_k^q) = [r_k, \dot{r}_k, \theta_k]^T, \quad (9)$$

where

$$\begin{cases} r_k = \sqrt{x_k^2 + y_k^2} \\ \dot{r}_k = (\dot{x}_k x_k + \dot{y}_k y_k) / r_k \\ \theta_k = \arctan(y_k / x_k) \end{cases}, \quad (10)$$

i.e., the range, velocity, and azimuth angle, respectively. $\mathbf{v}_k^q \sim \mathcal{N}(\mathbf{0}, \mathbf{R}_k^q)$, and

$$\mathbf{R}_k^q = \text{diag}(\sigma_{r_k^q}^2, \sigma_{\dot{r}_k^q}^2, \sigma_{\theta_k^q}^2), \quad (11)$$

where $\sigma_{r_k^q}^2$, $\sigma_{\dot{r}_k^q}^2$, and $\sigma_{\theta_k^q}^2$ are the CRLBs of the estimate accuracy of range, velocity, and azimuth angle, respectively

$$\begin{cases} \sigma_{r_k^q}^2 \propto \left(\alpha_k^q P_k^q |h_k^q|^2 |\beta_k^q|^2 \right)^{-1} \\ \sigma_{\dot{r}_k^q}^2 \propto \left(\alpha_k^q P_k^q |h_k^q|^2 |T_k^q|^2 \right)^{-1} \\ \sigma_{\theta_k^q}^2 \propto \left(\alpha_k^q P_k^q |h_k^q|^2 / B_w \right)^{-1} \end{cases}, \quad (12)$$

where $\propto$ is the direct proportion symbol, and B_w is the null-to-null beam width of the receive antenna [23].

III. RELATIONSHIP BETWEEN TRANSMIT PARAMETERS AND PREDICTED PCRLB

The PCRLB gives a lower and tight bound for any unbiased estimator [29]. More importantly, it is predictive, which is a salient characteristic for the resource allocation in target tracking. Therefore, this section derives the PCRLB in terms of power and bandwidth to guide the their allocation.

If let $\hat{\mathbf{x}}_k^q(\mathbf{z}_k^q)$ is an unbiased estimate of the target state, we will have the following Bayesian Cramer-Rao inequality [29]

$$\mathbb{E}_{\mathbf{x}_k^q, \mathbf{z}_k^q} \left[(\hat{\mathbf{x}}_k^q(\mathbf{z}_k^q) - \mathbf{x}_k^q) (\hat{\mathbf{x}}_k^q(\mathbf{z}_k^q) - \mathbf{x}_k^q)^T \right] \geq \mathbf{J}^{-1}(\mathbf{x}_k^q), \quad (13)$$

where $\mathbb{E}_{\mathbf{x}_k^q, \mathbf{z}_k^q}$ is the expectation operation in terms of the target state and the measurement. $\mathbf{J}(\mathbf{x}_k^q)$ is the Bayesian Fisher information matrix (BFIM), i.e., the inverse matrix of PCRLB, and it can be obtained by

$$\mathbf{J}(\mathbf{x}_k^q) = -\mathbb{E}_{\mathbf{x}_k^q, \mathbf{z}_k^q} \left[\nabla_{\mathbf{x}_k^q}^T \ln p(\mathbf{z}_k^q | \mathbf{x}_k^q) \right], \quad (14)$$

where the notation $\nabla_{\eta}^{\kappa} = \nabla_{\eta} \nabla_{\kappa}^T$ is the second order partial derivative vectors. The joint PDF in (14) can be factorized as

$$p(\mathbf{z}_k^q, \mathbf{x}_k^q) = p(\mathbf{x}_k^q) p(\mathbf{z}_k^q | \mathbf{x}_k^q), \quad (15)$$

where $p(\mathbf{z}_k^q, \mathbf{x}_k^q)$ is the joint PDF of $(\mathbf{z}_k^q, \mathbf{x}_k^q)$, $p(\mathbf{x}_k^q)$ is the PDF of the target state, and $p(\mathbf{z}_k^q | \mathbf{x}_k^q)$ is the joint conditional PDF. By blocking BFIM, [29] purposes an elegant recursive method for the discrete-time nonlinear filtering, which avoids the manipulation on the large matrix

$$\mathbf{J}(\mathbf{x}_k^q) = \mathbf{J}_P(\mathbf{x}_k^q) + \mathbf{J}_D(\mathbf{x}_k^q), \quad (16)$$

where $\mathbf{J}_P(\mathbf{x}_k^q)$ and $\mathbf{J}_D(\mathbf{x}_k^q)$ are the prior information FIM and data FIM, respectively. They are calculated as

$$\begin{cases} \mathbf{J}_P(\mathbf{x}_k^q) = -\mathbb{E}_{\mathbf{x}_k^q} \left[\nabla_{\mathbf{x}_k^q}^T \ln p(\mathbf{x}_k^q) \right] \\ \quad = \left[\mathbf{D}_{k-1}^{22} - \mathbf{D}_{k-1}^{21} (\mathbf{J}(\mathbf{x}_{k-1}^q) + \mathbf{D}_{k-1}^{11})^{-1} \mathbf{D}_{k-1}^{12} \right]^{-1}, \\ \mathbf{J}_D(\mathbf{x}_k^q) = -\mathbb{E}_{\mathbf{x}_k^q, \mathbf{z}_k^q} \left[\nabla_{\mathbf{x}_k^q}^T \ln p(\mathbf{z}_k^q | \mathbf{x}_k^q) \right] \end{cases}, \quad (17)$$

where

$$\begin{cases} \mathbf{D}_{k-1}^{11} = -\mathbb{E}_{\mathbf{x}_{k-1}^q, \mathbf{x}_k^q} \left[\nabla_{\mathbf{x}_{k-1}^q}^T \ln p(\mathbf{x}_k^q | \mathbf{x}_{k-1}^q) \right] \\ \mathbf{D}_{k-1}^{12} = -\mathbb{E}_{\mathbf{x}_{k-1}^q, \mathbf{x}_k^q} \left[\nabla_{\mathbf{x}_{k-1}^q}^T \ln p(\mathbf{x}_k^q | \mathbf{x}_{k-1}^q) \right] \\ \mathbf{D}_{k-1}^{22} = -\mathbb{E}_{\mathbf{x}_{k-1}^q, \mathbf{x}_k^q} \left[\nabla_{\mathbf{x}_k^q}^T \ln p(\mathbf{x}_k^q | \mathbf{x}_{k-1}^q) \right] \end{cases} = (\mathbf{D}_{k-1}^{21})^T. \quad (18)$$

For the target dynamics following the Gaussian distribution, which is described in (5), the three terms in (18) can be easily obtained by dropping out the expectation operator

$$\begin{cases} \mathbf{D}_{k-1}^{11} = \mathbf{F}_q^T \mathbf{Q}_q^{-1} \mathbf{F}_q \\ \mathbf{D}_{k-1}^{12} = -\mathbf{F}_q^T \mathbf{Q}_q^{-1} \\ \mathbf{D}_{k-1}^{22} = \mathbf{Q}_q^{-1} \end{cases}. \quad (19)$$

Substituting (19) into the first term of (17), and using the matrix inverse lemma, $\mathbf{J}_P(\mathbf{x}_k^q)$ is obtained as:

$$\mathbf{J}_P(\mathbf{x}_k^q) = (\mathbf{Q}_q + \mathbf{F}_q \mathbf{J}^{-1}(\mathbf{x}_{k-1}^q) \mathbf{F}_q^T)^{-1}. \quad (20)$$

As to $\mathbf{J}_D(\mathbf{x}_k^q)$, we have the following equation after substituting (15) into (17):

$$\begin{aligned} \mathbf{J}_D(\mathbf{x}_k^q) &= -\mathbb{E}_{\mathbf{x}_k^q, \mathbf{z}_k^q} \left[\nabla_{\mathbf{x}_k^q}^T \ln p(\mathbf{z}_k^q | \mathbf{x}_k^q) \right] \\ &= \mathbb{E}_{\mathbf{x}_k} \left\{ -\mathbb{E}_{\mathbf{z}_k^q | \mathbf{x}_k^q} \left[\nabla_{\mathbf{x}_k^q}^T \ln p(\mathbf{z}_k^q | \mathbf{x}_k^q) \right] \right\} \\ &= \mathbb{E}_{\mathbf{x}_k^q} \left[(\mathbf{H}_k^q)^T (\mathbf{R}_k^q)^{-1} \mathbf{H}_k^q \right], \end{aligned} \quad (21)$$

where $\mathbf{H}_k^q$ is the Jacobian matrix of the measurement function $h(\cdot)$ with respect to the target state $\mathbf{x}_k^q$. Obviously, the prior information FIM is only dependent on the target dynamics, and the allocated power and bandwidth have effects on $\mathbf{R}_k^q$, and thus on the PCRLB. However, the expectation operation in the data FIM means that, to achieve the PCRLB, the Monte Carlo techniques should be implemented. That will lead to a high computational cost. To make it more practical, a small process noise is assumed to approximate the BFIM to [23]

$$\begin{aligned} \mathbf{J}(\mathbf{x}_k^q) &= (\mathbf{Q}_q + \mathbf{F}_q \mathbf{J}^{-1}(\mathbf{x}_{k-1}^q) \mathbf{F}_q^T)^{-1} \\ &\quad + (\mathbf{H}_k^q)^T (\mathbf{R}_k^q)^{-1} \mathbf{H}_k^q \Big|_{\mathbf{x}_k^q = \hat{\mathbf{x}}_{k|k-1}^q} \\ &= (\mathbf{Q}_q + \mathbf{F}_q \mathbf{J}^{-1}(\mathbf{x}_{k-1}^q) \mathbf{F}_q^T)^{-1} + (\hat{\mathbf{H}}_k^q)^T (\hat{\mathbf{R}}_k^q)^{-1} \hat{\mathbf{H}}_k^q, \end{aligned} \quad (22)$$

where $\hat{\mathbf{x}}_{k|k-1}^q$ is the predicted target state at the $(k-1)$ -th sample interval. That means, we can predict the PCRLB at the next time based on the obtained target information at the current time.

By integrating the power and the bandwidth to be allocated, the BFIM is rewritten as:

$$\begin{aligned} \mathbf{J}(\mathbf{x}_k^q, P_k^q, \beta_k^q) &= (\mathbf{Q}_q + \mathbf{F}_q \mathbf{J}^{-1}(\mathbf{x}_{k-1}^q) \mathbf{F}_q^T)^{-1} \\ &\quad + (\hat{\mathbf{H}}_k^q)^T (\hat{\mathbf{R}}_k^q(P_k^q, \beta_k^q))^{-1} \hat{\mathbf{H}}_k^q. \end{aligned} \quad (23)$$

To remind the reader, the relationship between P_k^q , β_k^q and $\mathbf{R}_k^q$ is shown in (12). Considering the position error dominates the target state estimate, we extract the position submatrix of PCRLB as the metric. Thereby, we can focus on the power and bandwidth resources to be allocated. Additionally, since the position, velocity, and acceleration have heterogenous units, the trace of whole PCRLB may possess no meaning. By using the extraction, this situation can be avoided.

For the future use, we write the PCRLB of position as

$$\mathbb{G}(P_k^q, \beta_k^q) = \text{tr}(\mathbf{J}^{-1}(\mathbf{x}_k^q, P_k^q, \beta_k^q)). \quad (24)$$

where $\text{tr}(\cdot)$ is the trace operation.

IV. JOINT POWER AND BANDWIDTH ALLOCATION

A. Optimization Formulation

Mathematically, the optimal power and bandwidth allocation can be described as an optimization problem constrained by the limited resource budget. The PCRLB derived in section III is a useful measure to form the optimization model.

Assume that the power and the bandwidth allocation vectors are $\mathbf{P}_k = [P_k^1, P_k^2, \dots, P_k^Q]^T$ and $\beta_k = [\beta_k^1, \beta_k^2, \dots, \beta_k^Q]^T$, respectively. Our aim is to jointly optimize the two resources to achieve the minimum of target state estimate errors. In previous works, the sum or the maximum of PCRLB [21]–[24] is usually adopted as the objective function, which may be oversimplified. To describe the specific requirement of tracking accuracy of each target, we introduce a weight factor, and the sum of weighted PCRLBs is used to perform this task

$$\min_{\mathbf{P}_k, \beta_k} \mathbb{F}(\mathbf{P}_k, \beta_k) = \min_{\mathbf{P}_k, \beta_k} \sum_{q=1}^Q \chi_k^q G(P_k^q, \beta_k^q), \quad (25)$$

where χ_k^q represents the importance of the q -th target, and it can be determined by the comprehensive consideration on the position, velocity and other parameters [35]. However, detailed discussion on χ_k^q is out of this paper's scope. Just for simplicity, we will preset χ_k^q directly in the simulations. Then, the problem is formulated as

$$\begin{aligned} \min_{\mathbf{P}_k, \beta_k} \quad & \mathbb{F}(\mathbf{P}_k, \beta_k) \\ \text{s.t.} \quad & P_{\min} \leq P_k^q \leq P_{\max} \\ & \beta_{\min} \leq \beta_k^q \leq \beta_{\max} \\ & \mathbf{1}_Q^T \mathbf{P}_k = P_{\text{total}} \\ & \mathbf{1}_Q^T \beta_k = \beta_{\text{total}}. \end{aligned} \quad (26)$$

where $\mathbf{1}_Q^T = [1, 1, \dots, 1]_{1 \times Q}$. Note that when the allocated power is tuned, the probability of detection P_d will be changed, correspondingly. Therefore, we set the minimum threshold as $P_{\min}$ to meet the requirement of $P_d \geq 1$. $P_{\min}$ and $P_{\max}$ form an interval to bound P_k^q , and $\beta_{\min}$ and $\beta_{\max}$ can be handled accordingly. P_{total} and β_{total} are the maximum power and bandwidth that the system can provide at the k -th sample interval, separately.

B. Solution Algorithm

The detailed analysis of the objective function (25) is given in Appendix, which shows that the convexity of (26) is uncertain, though the constraints in (26) are linear and convex. In addition, it can be seen from (37) that the bandwidth has a quadratic form in the (25), and the power and the bandwidth variables are coupled in the objective function. The above two points leads to an intractable problem. In order to solve it efficiently, the cyclic minimizer [30] is adopted. Specially to say, the proposed solution algorithm tries to optimize one variable on the condition of another being fixed in a cyclical framework. The flow chart of the algorithm is shown in Fig. 2.

The steps of solution algorithm are described as follows.

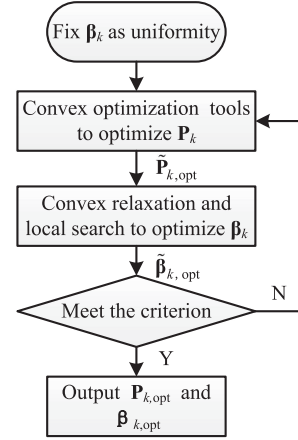


Fig. 2. Flowchart of the solution algorithm.

TABLE II
MODIFIED ZOUTENDIJK FEASIBLE DIRECTION METHOD

(1) Reformulation. The optimization model is normalized as: $\min f(\mathbf{x})$ $\text{s.t. } \mathbf{Ax} \geq \mathbf{b}$ $\mathbf{Ex} = \mathbf{e}$ then, input a feasible starting point $\mathbf{x}^{(l)}$, set $l=1$. (2) Set $\mathbf{I} = \{i g_i(\mathbf{x}^{(l)}) = 0\}$, and solve the following linear programming model: $\min c$ $\text{s.t. } \nabla f(\mathbf{x}^{(l)})^T \mathbf{d} - c \leq 0$ $\nabla g_i(\mathbf{x}^{(l)})^T \mathbf{d} + c \geq g_i(\mathbf{x}^{(l)}), i=1, 2, \dots, m$ $-1 \leq d_j \leq 1, j=1, 2, \dots, n$ then, obtain the optimal solution $(c^{(l)}, \mathbf{d}^{(l)})$. (3) If $c^{(l)} = 0$, stops and output the Fritz John point $\mathbf{x}^{(l)}$; otherwise, go to (4). (4) Solve $\min f(\mathbf{x}^{(l)} + \lambda \mathbf{d}^{(l)})$ $\text{s.t. } 0 \leq \lambda \leq \lambda_{\max}$ $\lambda_{\max} = \sup \{\lambda g_i(\mathbf{x}^{(l)} + \lambda \mathbf{d}^{(l)}) \geq 0, i=1, 2, \dots, m\}$ and obtain the optimal $\lambda^{(l)}$. (5) Let $\mathbf{x}^{(l+1)} = \mathbf{x}^{(l)} + \lambda^{(l)} \mathbf{d}^{(l)}$, and $l=l+1$, return to (2)
--

Step 1: Initialization. Set the bandwidth allocation as the uniformity, and go to Step 2.

Step 2: On the condition of the fixed bandwidth allocation, reformulate (26) as

$$\begin{aligned} \min \quad & \mathbb{F}(\mathbf{P}_k) \\ \text{s.t.} \quad & P_{\min} \leq P_k^q \leq P_{\max} \\ & \mathbf{1}_Q^T \mathbf{P}_k = P_{\text{total}}. \end{aligned} \quad (27)$$

Equation (27) shows a convex problem. That is because, first, the constraints are linear and convex. In addition, according to (38) in Appendix, the objective function is also convex. In this case, many standard convex optimization methods are applicable, e. g., the modified Zoutendijk feasible direction method [22], which is presented in Table II. Thus, the temporary power allocation vector $\hat{\mathbf{P}}_{k, \text{opt}}$ is easily obtainable.

TABLE III
LOCAL SEARCH ALGORITHM FOR (29)

(1) Initialization: $\beta_{l0} = \bar{\beta}_{k,\text{opt}}$, $\mathbb{F}(\bar{\mathbf{P}}_{k,\text{opt}}, \beta_{l0})$, iteration step $\Delta\beta_0$ and stop criterion ε .
(2) Repeat $\beta_l = \arg \max \left\{ \mathbb{F}(\bar{\mathbf{P}}_{k,\text{opt}}, \beta_{l-1}^+) - \mathbb{F}(\bar{\mathbf{P}}_{k,\text{opt}}, \beta_{l-1}) \right\}$ s.t. $\beta_{\min} \leq \beta_l^q \leq \beta_{\max}$ $\Delta\beta_l = \Delta\beta_{l-1} (\mathbf{1}_Q^T \beta_l / \mathbf{1}_Q^T \beta_{l-1})$
(3) while $\mathbb{F}(\bar{\mathbf{P}}_{k,\text{opt}}, \beta_{l-1}) - \mathbb{F}(\bar{\mathbf{P}}_{k,\text{opt}}, \beta_l) \geq \varepsilon$ set $\beta_l = \beta_{l-1}$, end

Step 3: Fix the power allocation as $\bar{\mathbf{P}}_{k,\text{opt}}$, and reformulate (26) as

$$\begin{aligned} \min_{\beta_k} \quad & \mathbb{F}(\beta_k) \\ \text{s.t.} \quad & \beta_{\min} \leq \beta_k^q \leq \beta_{\max} \\ & \mathbf{1}_Q^T \beta_k = \beta_{\text{total}}. \end{aligned} \quad (28)$$

As analyzed in Appendix, (28) is not always a convex problem, especially near the point zero. However, notice that the quadratic term of the bandwidth $|\beta_k^q|^2$ and the power term P_k^q can be exchanged in objective function (see (37)), we relax (28) by introducing a new vector $\gamma_k = [\gamma_k^1, \gamma_k^2, \dots, \gamma_k^Q]^T = [|\beta_k^1|^2, |\beta_k^2|^2, \dots, |\beta_k^Q|^2]^T$ and a factor λ_1 to reformulate the optimization problem (28) as

$$\begin{aligned} \min_{\gamma_k} \quad & \mathbb{F}(\gamma_k) \\ \text{s.t.} \quad & |\beta_{\min}|^2 \leq \gamma_k^q \leq |\beta_{\max}|^2 \\ & \beta_{\text{total}}^2 / Q \leq \mathbf{1}_Q^T \gamma_k \leq \lambda_1 \beta_{\text{total}}^2, \end{aligned} \quad (29)$$

where λ_1 satisfies

$$\lambda_1 \leq 1 - Q(Q-1) \beta_{\min}^2 / \beta_{\text{total}}^2. \quad (30)$$

Its value can be adjusted according to the simulation results. Specifically, we can set $\Delta\beta$ as a step size to increase or decrease λ_1 according to the output of the objective function. We note that the aim of such an adjustment is to obtain more accurate solutions, and it will not consume too much time. Similar to (27), (29) is a convex optimization problem. As such, the modified Zoutendijk feasible direction method can be applied again.

Step 4: Due to the relaxation in (29), the solution may be not optimal. Thus, the solution $\bar{\beta}_{k,\text{opt}}$ to (29) is then adopted as the initial point for a local search algorithm, which is shown in Table III, to improve the solution precision. At the l -th iteration, we reduce one of elements in $\bar{\beta}_{k,\text{opt}}$ by $\Delta\beta_l$, and add this amount to any of other elements, to generate a vector β_l^+ . This single change that offers the highest gain in reducing the PCRLB is used as the initial solution for the next iteration. The process stops until the deduction of PCRLB in one iteration to another is satisfied the stopping condition ε . Finally, output the temporary bandwidth allocation vector $\bar{\beta}_{k,\text{opt}}$.

Step 5: Return to Step 2, and stops until the objective function converges, i.e., $|\mathbb{F}_{j+1}(\bar{\mathbf{P}}_{k,\text{opt}}, \bar{\beta}_{k,\text{opt}}) - \mathbb{F}_j(\bar{\mathbf{P}}_{k,\text{opt}}, \bar{\beta}_{k,\text{opt}})| \leq \varepsilon_0$.

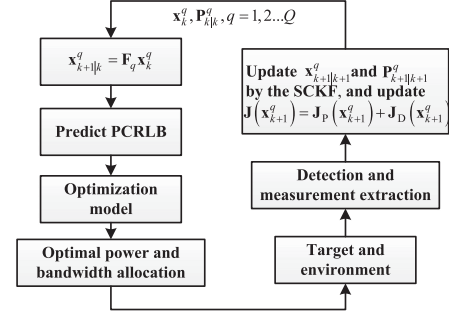


Fig. 3. Flowchart of the closed loop tracking system.

$|\mathbb{F}_j(\bar{\mathbf{P}}_{k,\text{opt}}, \bar{\beta}_{k,\text{opt}}) - \mathbb{F}_{j-1}(\bar{\mathbf{P}}_{k,\text{opt}}, \bar{\beta}_{k,\text{opt}})| \leq \varepsilon_0$, where $\mathbb{F}_j(\bar{\mathbf{P}}_{k,\text{opt}}, \bar{\beta}_{k,\text{opt}})$ is the objective function value obtained from the j -th cycle, and ε_0 is the preset threshold. Then, obtain $\mathbf{P}_{k,\text{opt}}$ and $\beta_{k,\text{opt}}$.

Above all, through the convex relaxation and the cyclic minimizer, the joint power and bandwidth allocation problem can be solved efficiently. It is noticeable that the optimization order can be swapped in the proposed solver, i.e., first fix the power and optimize the bandwidth, and then fix the bandwidth and optimize the power. However, both the performance and the runtime will not be influenced too much. More details will be shown in Section V-E.

C. Computational Complexity

In this subsection, the computational complexity of the proposed solution algorithm is evaluated. Our algorithm involves the Modified Zoutendijk feasible direction method and the local search algorithm in a cycle manner. For the Modified Zoutendijk feasible direction method, it requires at most $\mathcal{O}(\sqrt{Q})$ iterations to achieve the given accuracy. The complexity of the matrix inverse operation is $\mathcal{O}(Q^3)$ in each iteration. In the local search, $\mathcal{O}(Q)$ iterations are required in each round. The complexity of matrix inverse also should be considered. Thus, we can conclude that the proposed solution algorithm is with the total complexity of $\mathcal{O}(Q^4)$.

D. Closed Loop Tracking System

To tackle the nonlinear measurement from the radar coordinate, we choose the SCKF [32] to obtain the accurate target state estimates, because it possesses the merits of low structure, high precision and numerical stability [33]. After acquiring the target information at the current time, the predicted PCRLBs are used as the optimization criterion to serve the optimal power and bandwidth allocation for the next time. Then, the optimal resource allocation results in the tracking accuracy change, and the actual PCRLBs are updated according to the actual target state estimates at the next time. Finally, the PCRLBs are predicted again, which renders a closed loop tracking system. The flowchart is presented in Fig. 3.

V. SIMULATIONS AND ANALYSIS

In this section, some numerical results are provided to demonstrate the effectiveness of the proposed method. A collocated MIMO radar located at the origin point is chosen for analysis.

TABLE IV
 TARGET PARAMETERS

Target index	1	2	3
Initial position ($\times 10^5$ m)	(4.83, -0.11)	(2.73, 0.76)	(2.09, 0.59)
Initial velocity (m/s)	(-590, -200)	(-885, -400)	(-1280, -990)
Initial acceleration (m/s^2)	(-10, 0)	(-15, 0)	(-20, -10)

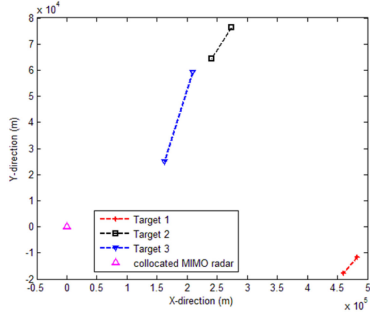


Fig. 4. Target trajectories with respect to the MIMO radar.

The total transmit power is $P_{\text{total}} = 10$ kW. Each transmit element transmits the orthogonal frequency division multiplexing-linear frequency modulation (OFDM-LFM) signal. The effective time width is $T_{\text{total}} = 0.01\text{ms}$, and the total effective bandwidth is $\beta_{\text{total}} = 100$ MHz. There are $Q = 3$ widely separated targets, whose parameters are shown in Table IV. The geometry positions of targets with respect to the MIMO radar is presented in Fig. 4. The transmit bandwidth and the power are restricted within $[0.1\beta_{\text{total}}, 0.8\beta_{\text{total}}]$ and $[0.1P_{\text{total}}, 0.8P_{\text{total}}]$, respectively. A sequence of 30 frames with sample interval $T_s = 1$ s are used in each Monte Carlo trial. The standard measurement error is assumed as: $\mathbf{R} = \text{diag}(\sigma_r^2, \sigma_{\dot{r}}^2, \sigma_\theta^2)$, where $\sigma_r = 100$ m, $\sigma_{\dot{r}} = 10$ m/s and $\sigma_\theta = 0.1$ rad, which is obtained from the range of 500 km using $0.1P_{\text{total}}$ and $0.1\beta_{\text{total}}$ to illuminate the target with RCS $|h_k^q| = 1\text{m}^2$. Three benchmarks are utilized to compare against the proposed algorithm: both uniform allocation (Both uniform), optimal power with uniform bandwidth allocation (Power optimal), and optimal bandwidth with uniform power allocation (Bandwidth optimal). For the fair comparison, at the initial time, each method allocates the two resources uniformly.

To investigate the factors that influence the resource allocation and the robustness of the proposed method, four cases are assumed. The sum of weighted PCRLBs in terms of position is used as the evaluation metric, as shown in (25). The corresponding tracking root mean square error (RMSE) at the k -th sample interval is defined as:

$$\text{RMSE}_k$$

$$= \sum_{q=1}^Q \chi_k^q \sqrt{\frac{1}{N_{\text{sim}}} \sum_{j=1}^{N_{\text{sim}}} \left((x_k^q - \hat{x}_{k,j}^q)^2 + (y_k^q - \hat{y}_{k,j}^q)^2 \right)}, \quad (31)$$

where $N_{\text{sim}} = 100$ is the number of Monte Carlo trials, and $(\hat{x}_{k,j}^q, \hat{y}_{k,j}^q)$ denotes the position estimate for the q -th target in

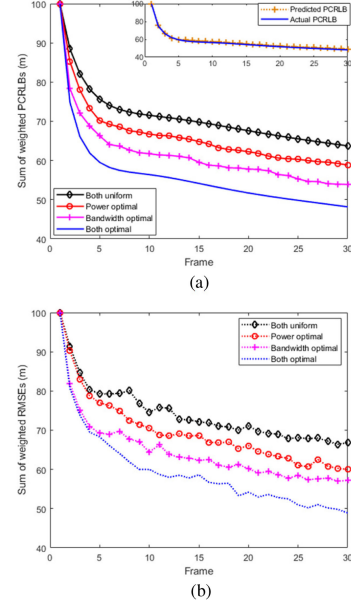


Fig. 5. Performance in terms of PCRLB and RMSE in case 1. (a) PCRLB comparison. (b) RMSE comparison.

the j -th trail. Our simulations are run in MATLAB 2014a on a computer with 3.7 GHz CPU and 8GB RAM.

A. Case 1: Effect of Distance

In this case, $\chi_k^q = 1/3$, for $q = 1, 2, 3$, and $k = 1, 2, \dots, 30$, all the target RCSs are equal to one, and their model errors are $\kappa_q = 1$, for $q = 1, 2, 3$. (κ_q is the process noise intensity, which is shown in (7).) Therefore, the resource allocation only depends on the deployment of targets.

The subfigure in Fig. 5(a) shows that the predicted PCRLB is very close to the actual PCRLB. This finding verifies the correctness of the established optimization model. Fig. 5(a) and Fig. 5(b) compare the sum of weighted PCRLBs as well as RMSEs in terms of position, respectively. Obviously, the joint power and bandwidth allocation performs the best among all the algorithms. By contrast, the uniform allocation has the worst performance. This is because the proposed strategy utilizes the feedback information to guide the resource allocation, rendering the minimum estimate errors at each frame. Moreover, we can see that the optimal bandwidth allocation has more effect on the performance improvement compared with the optimal power allocation. This can be interpreted by (37), where the bandwidth has a quadratic form in the position submatrix of PCRLB.

Fig. 6 shows the power and the bandwidth allocation achieved by the proposed method, where the color in each rectangle represents the ratio of allocated power

$$r_{\text{power},k}^q = \frac{P_k^q}{P_{\text{total}}}, \quad (32)$$

and the ratio of allocated bandwidth

$$r_{\text{bandwidth},k}^q = \frac{\beta_k^q}{\beta_{\text{total}}}. \quad (33)$$

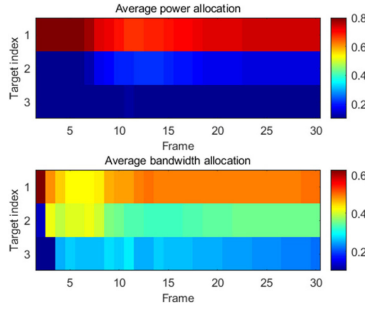


Fig. 6. Average resource allocation in case 1.

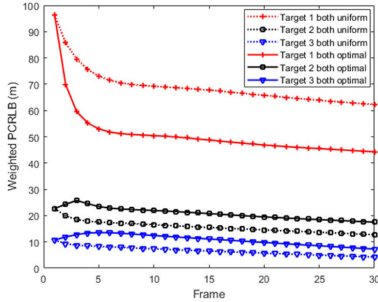


Fig. 7. PCRLB of each target in case 1.

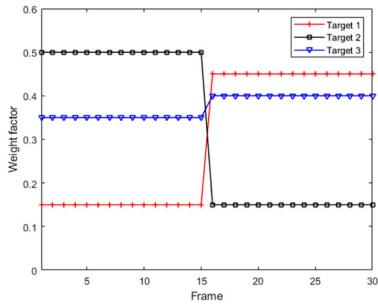


Fig. 8. Time-varying weight factor model.

In Fig. 6, the cyaneous color denotes that the ratio is zero, while the crimson color denotes that the ratio achieves the maximum. It can be seen that more power and bandwidth are allocated to the target with farther distance, i.e., target 1. This result is similar to many previous works [23], [24] that adopt the maximum PCRLB as the cost function. (Note that refs. [23], [24] only study the power allocation, but this paper investigates the joint power and bandwidth allocation.) However, the evident divergence between the PCRLBs after the resource allocation, shown in Fig. 7, should be highlighted. This point is totally different from [23], [24], where the PCRLBs converge after the resource allocation. In the next subsection, we will show the flexibility of the established objective function.

B. Case 2: Effect of Weight

In this case, χ_k^q is assumed to be a time-varying value, as shown in Fig. 8. All the target RCSs are equal to one, and their model errors are $\kappa_q = 1$, for $q = 1, 2, 3$. Therefore, we can see the effect of weight factor on the resource allocation.

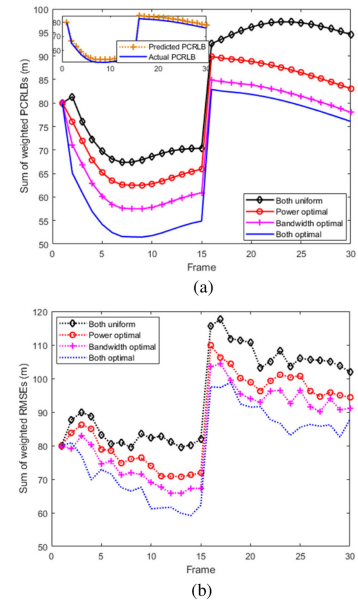


Fig. 9. Performance in terms of PCRLB and RMSE in case 2. (a) PCRLB comparison. (b) RMSE comparison.

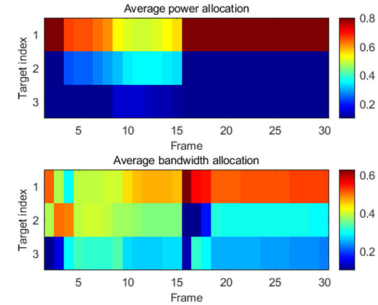


Fig. 10. Average resource allocation in case 2.

Fig. 9(a) and (b) show the performance comparison among all the algorithms. It can be seen that the uniform allocation regard the targets as the equal importance, and thus, the highest estimate errors are provided. By contrast, the proposed strategy considers the difference of target importance, and achieves the lowest PCRLB and RMSE.

The resource allocation results of the proposed strategy are shown in Fig 10. In this case, target 2 and target 3 are given more resource before 16-th frame compared with case 1, which is because the two targets are given higher weight factors. In Fig. 11, we can see that the estimate accuracy of target 1 is improved more evidently after 16-th frame. This can be interpreted as the same reason, where target 1 is given lower weight before the 16-th frame. These points show the flexibility of the established optimization model, where the target importance can be elegantly tuned to guide the resource allocation.

C. Case 3: Effect of RCS

In this case, we consider the time-varying RCS model, which is shown in Fig. 12. $\chi_k^q = 1/3$, for $q = 1, 2, 3$, and $k = 1, 2, \dots, 30$, and the model errors of all the targets are equal to 1. As

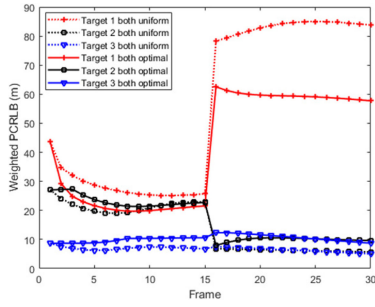


Fig. 11. PCRLB of each target in case 3.

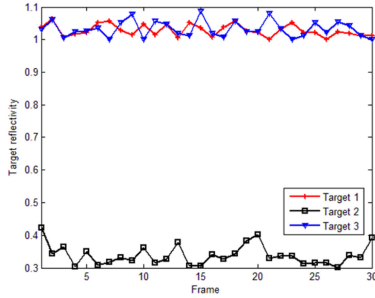


Fig. 12. Time-varying target RCS model.

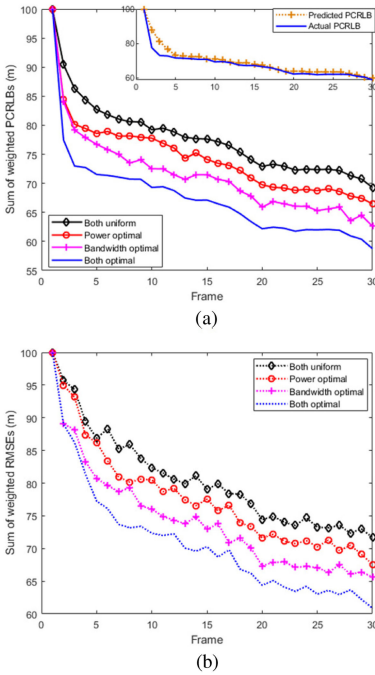


Fig. 13. Performance in terms of PCRLB and RMSE in case 3. (a) PCRLB comparison. (b) RMSE comparison.

such, the results can support the influence of target RCS on the resource allocation.

Fig. 13 gives the performance comparison of the four algorithms, which proves that the joint power and bandwidth allocation can make more full use of the limited resources to track multiple targets. Moreover, the subfigure in Fig. 13 shows the closeness of the predicted PCRLB and the actual PCRLB. Such a

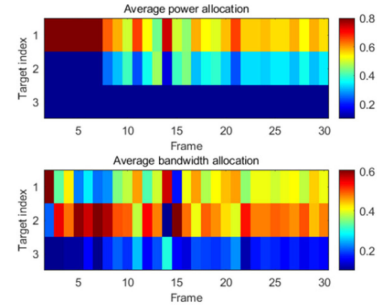


Fig. 14. Average resource allocation in case 3.

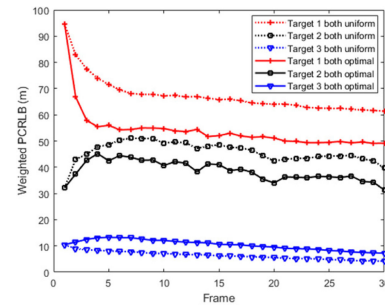


Fig. 15. PCRLB of each target in case 3.

result also indicates the validation of the proposed optimization model.

Figs. 14 and 15 illustrate the effect of target RCS on the resource allocation and the weighted PCRLB of each target, respectively. In this case, target 2 is obtained more power and bandwidth resource compared with case 1, which is due to its lowest target reflectivity. Target 3 still gains the least resources, owing to its closest distance as well as the strong RCS. Because of the same reason, the PCRLB of target 2 decreases after the resource allocation, which is contrast to case 1.

D. Case 4: Effect of Model Error

In this case, we will investigate the effect of model errors on the resource allocation. Two dynamic models are considered, where $\kappa_q = 1$, for $q = 1, 3$, and $\kappa_q = 1000$, for $q = 2$. $\chi_k^q = 1/3$, for $q = 1, 2, 3$, and $k = 1, 2, \dots, 30$, and the target RCSs are all equal to 1.

The subfigure in Fig. 16(a) shows a fact that the gap between the predicted PCRLB and the actual PCRLB is a little wider than the former three cases, which is due to the larger model error of target 2. However, such a gap has little influence on the proposed method. From Fig. 16(a) and (b), we know that the superiority of the joint power and bandwidth allocation strategy still holds in this case.

The corresponding resource allocation and the weighted PCRLB of each target are given in Figs. 17 and 18, respectively. Since target 2 possesses the worst dynamic model, more resource is given to it compared with case 1, which results in an evident lower PCRLB.

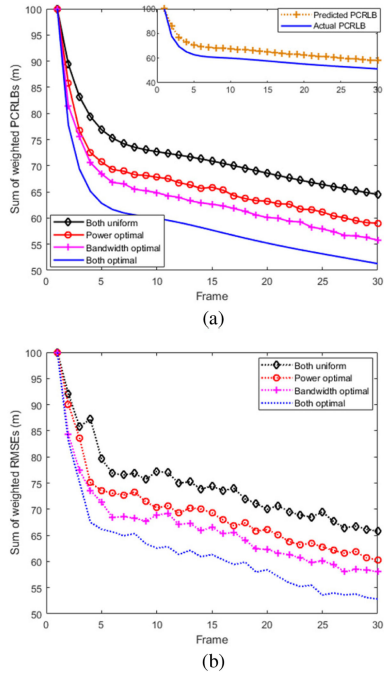


Fig. 16. Performance in terms of PCRLB and RMSE in case 4. (a) PCRLB comparison. (b) RMSE comparison.

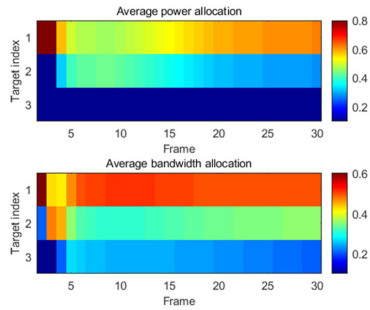


Fig. 17. Average resource allocation in case 4.

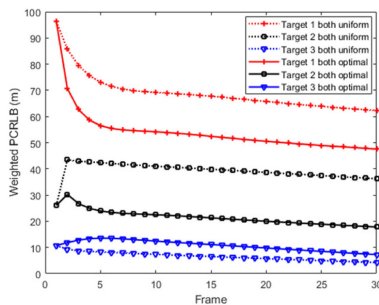


Fig. 18. PCRLB of each target in case 4.

E. Efficiency of the Proposed Solution Algorithm

To show the efficiency of the proposed solution algorithm, we use the genetic algorithm as the benchmark for comparison. The performance is presented in Fig. 19.

Fig. 19(a) shows the PCRLBs achieved by the two algorithms in case 1. Evidently, the tracking performance of the proposed

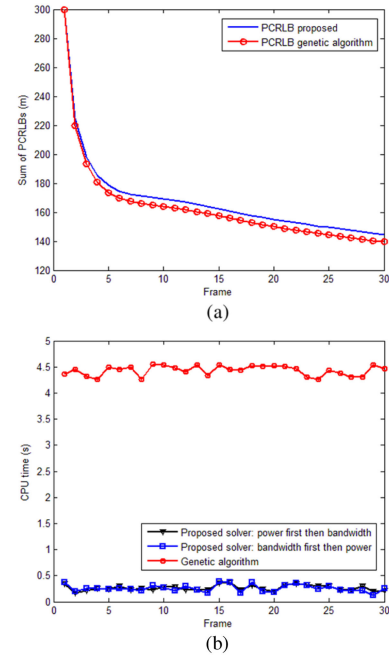


Fig. 19. Performance comparison between the proposed solver and genetic algorithm. (a) PCRLB. (b) Average runtime.

strategy is very close to the genetic algorithm. Fig. 19(b) shows the CPU time of the genetic algorithm and two variants of the proposed solution algorithm, i.e., first optimize power then bandwidth and first optimize bandwidth then power. It can be seen that the genetic algorithm has larger computational complexity. By contrast, the proposed solver is very efficient in solving the optimization model. In addition, the runtime between two variants of proposed solver is very small. (We note that the PCRLBs achieved by the two variants are the same.) The results indicate the effectiveness and efficiency of the proposed solver. Actually, the iteration of the cycle minimizer is always less than ten times, since the solution accuracy in the power optimization and bandwidth optimization is well controlled by the Modified Zoutendijk feasible direction method and the local search algorithm.

From the cases above, some conclusions are drawn as follows.

- 1) The proposed strategy could effectively improve the sum of weighted PCRLBs, and it shows adaptivity and robustness when faced with different cases.
- 2) In the proposed strategy, the target with farther distance, higher weight factor, lower target reflectivity and worse model error will be allocated more power and bandwidth resources.
- 3) The bandwidth allocation has more effect on the PCRLB improvement compared with the power allocation.

VI. CONCLUSION

In this paper, we proposed a joint power and bandwidth allocation strategy for collocated MIMO radar tracking multiple targets with the simultaneous multi-beam mode. The optimization model is formulated with the aim of minimizing the sum of

weighted PCRLBs of multiple targets, whose prior information is obtained in the tracking recursion cycle. By using the convex relaxation and the cycle minimizer, the intractable problem is transformed into a two-stage convex problem, and the solution is provided efficiently. The simulation results show the superior performance of the proposed strategy compared with three uniform resource allocation methods. Moreover, it shows adaptivity and robustness when faced with different cases.

We note that this work is based on condition of the perfect detection. Future works might concern that the measurement origins from false alarms and clutters [36], and the extension to large scale radar networks [37]. In addition, other approaches, such as the game theory [38] and the improved intelligence algorithm [39], [40] may be taken into account.

APPENDIX

The analysis of the objective function (25).

According to (12), we can extract P_k^q from $\mathbf{R}_k^q$ and rewrite (22) as

$$\mathbf{J}(\mathbf{x}_k^q, P_k^q) = (\mathbf{Q}_q + \mathbf{F}_q \mathbf{J}^{-1}(\mathbf{x}_{k-1}^q) \mathbf{F}_q^T)^{-1} + (\hat{\mathbf{H}}_k^q)^T (P_k^q \hat{\mathbf{Y}}_k^q)^{-1} \hat{\mathbf{H}}_k^q \quad (34)$$

where $\hat{\mathbf{Y}}_k^q$ is the remaining part after the extraction. Then, we define $g(P) = \text{tr}[(\mathbf{Z} + P\mathbf{V})^{-1}]$ according to (34) and (24), and restrict g to the interval of P where $\mathbf{Z} + P\mathbf{V} > 0$. An intuitive illustration is that $\mathbf{Z}$ is the prior information FIM (20), and $P\mathbf{V}$ is the data FIM (21). Then, we have

$$\begin{aligned} g(P) &= \text{tr}[(\mathbf{Z} + P\mathbf{V})^{-1}] \\ &= \text{tr}\left[\mathbf{Z}^{-1/2}(\mathbf{I} + P\mathbf{Z}^{-1/2}\mathbf{V}\mathbf{Z}^{-1/2})^{-1}\mathbf{Z}^{-1/2}\right] \\ &= \text{tr}\left[\mathbf{Z}^{-1/2}\mathbf{Q}(\mathbf{I} + P\mathbf{\Pi})^{-1}\mathbf{Q}^T\mathbf{Z}^{-1/2}\right] \\ &= \text{tr}\left[\mathbf{Q}^T\mathbf{Z}^{-1/2}\mathbf{Z}^{-1/2}\mathbf{Q}(\mathbf{I} + P\mathbf{\Pi})^{-1}\right] \\ &= \sum_{i=1}^6 \left(\mathbf{Q}^T\mathbf{Z}^{-1/2}\mathbf{Z}^{-1/2}\mathbf{Q}\right)_{ii} (1 + P\pi_i)^{-1} \end{aligned} \quad (35)$$

where $(\cdot)_{ii}$ is the i -th diagonal element in the matrix. The eigenvalue decomposition $\mathbf{Z}^{-1/2}\mathbf{V}\mathbf{Z}^{-1/2} = \mathbf{Q}\mathbf{\Pi}\mathbf{Q}^T$ is implied in (35), and π_i is the corresponding eigenvalues. Note that the position PCRLB is extracted as the performance metric, thus, the objective function in terms of the power allocation variable can be written as

$$\begin{aligned} \mathbb{G}(P) &= \sum_{i=1}^2 \left(\mathbf{Q}^T\mathbf{Z}^{-1/2}\mathbf{Z}^{-1/2}\mathbf{Q}\right)_{ii} (1 + P\pi_i)^{-1} \\ &= (a_1 + b_1 P)^{-1} + (a_2 + b_2 P)^{-1} \end{aligned} \quad (36)$$

where a_1, b_1, a_2 , and b_2 are all positive values due to the symmetric positive definite property of (20) and (21). Notice that the index q and k are omitted for brevity in (36). Accordingly,

(24) can be written as

$$\mathbb{G}(P, \beta) = (a_1 + c_1 P \beta^2)^{-1} + (a_2 + c_2 P \beta^2)^{-1} \quad (37)$$

where c_1 and c_2 are both positive values. The second and the mixed partial derivatives of (37) can be calculated as

$$\frac{\partial^2 \mathbb{G}(P, \beta)}{\partial P^2} = 2c_1^2 \beta^4 / (a_1 + c_1 P \beta^2)^3 + 2c_2^2 \beta^4 / (a_2 + c_2 P \beta^2)^3, \quad (38)$$

$$\begin{aligned} \frac{\partial^2 \mathbb{G}(P, \beta)}{\partial P \partial \beta} &= [8\beta^2 - 2(a_1 + c_1 P \beta^2)] / (a_1 + c_1 P \beta^2)^3 \\ &\quad + [8\beta^2 - 2(a_2 + c_2 P \beta^2)] / (a_2 + c_2 P \beta^2)^3, \end{aligned} \quad (39)$$

and

$$\begin{aligned} \frac{\partial^2 \mathbb{G}(P, \beta)}{\partial P \partial \beta} &= (2c_1^2 P \beta^3 - 2c_1 a_1 \beta) / (a_1 + c_1 P \beta^2)^3 \\ &\quad + (2c_2^2 P \beta^3 - 2c_2 a_2 \beta) / (a_2 + c_2 P \beta^2)^3, \end{aligned} \quad (40)$$

respectively. Obviously, $\partial^2 \mathbb{G}(P, \beta) / \partial P^2 \geq 0$, and we can say that the objective function (25) is convex in terms of the power allocation. However, we can't ascertain the positivity or negativity of $[\partial^2 \mathbb{G}(P, \beta) / \partial \beta^2] \times [\partial^2 \mathbb{G}(P, \beta) / \partial P^2] - [\partial^2 \mathbb{G}(P, \beta) / \partial P \partial \beta]^2$ due to the uncertainty of a_1, a_2, c_1 and c_2 . Thus, we can't ascertain the convexity of (26). However, according to (39), we can say that $\partial^2 \mathbb{G}(P, \beta) / \partial \beta^2 \leq 0$ near the point of $\beta = 0$. Such a characteristic is very important, because β is restricted within the interval $[0, 1]$ after the normalization. Thus, we can say that the objective function (25) is not always convex in terms of the bandwidth allocation.

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